

1.

```
SELECT emp_id, emp_name, dept_id, salary
FROM employees e
WHERE salary > (
    SELECT AVG(salary)
    FROM employees
    WHERE dept_id = e.dept_id
);
```

2.

```
SELECT emp_id, emp_name, dept_id, salary
FROM employees e
WHERE salary = (
    SELECT MAX(salary)
    FROM employees
    WHERE dept_id = e.dept_id
);
```

3.

```
SELECT MAX(salary) AS second_highest_salary
FROM employees
WHERE salary < (
    SELECT MAX(salary)
    FROM employees
);
```

4.

```
SELECT emp_name, salary
FROM employees
WHERE salary = (
    SELECT MAX(salary)
```

```
FROM employees
WHERE salary < (
    SELECT MAX(salary)
    FROM employees
)
);
```

5.

```
SELECT emp_id, emp_name, dept_id, salary
FROM employees e
WHERE manager_id IS NOT NULL
AND salary > (
    SELECT salary
    FROM employees m
    WHERE m.emp_id = e.manager_id
);
```

6.

```
SELECT emp_id, emp_name, salary
FROM employees e
WHERE EXISTS (
    SELECT 1
    FROM employees r
    WHERE r.manager_id = e.emp_id
    AND r.salary > e.salary
);
```

7.

```
SELECT dept_id
FROM employees
GROUP BY dept_id
```

```
HAVING AVG(salary) > (  
    SELECT AVG(salary)  
    FROM employees  
);
```

8.

```
SELECT emp_id, emp_name, dept_id, salary  
FROM employees  
WHERE dept_id IN (  
    SELECT dept_id  
    FROM employees  
    GROUP BY dept_id  
    HAVING AVG(salary) > (  
        SELECT AVG(salary)  
        FROM employees  
    )  
);
```

9.

```
SELECT emp_id, emp_name, salary  
FROM employees  
WHERE salary > ALL (  
    SELECT salary  
    FROM employees  
    WHERE dept_id = (  
        SELECT dept_id  
        FROM departments  
        WHERE dept_name = 'HR'  
    )  
);
```

10.

```
SELECT emp_id, emp_name, salary
FROM employees
WHERE salary > ANY (
    SELECT salary
    FROM employees
    WHERE dept_id = (
        SELECT dept_id
        FROM departments
        WHERE dept_name = 'Research'
    )
);
```

11.

```
SELECT dept_id, dept_name
FROM departments d
WHERE NOT EXISTS (
    SELECT 1
    FROM projects p
    WHERE p.dept_id = d.dept_id
);
```

12.

```
SELECT emp_id, emp_name, dept_id, salary
FROM employees e
WHERE (
    SELECT COUNT(*)
    FROM projects p
    WHERE p.dept_id = e.dept_id
) > 1;
```

13.

```
SELECT emp_id, emp_name, dept_id, salary
FROM employees
WHERE dept_id IN (
    SELECT dept_id
    FROM projects
    WHERE budget = (
        SELECT MAX(budget)
        FROM projects
    )
);
```

14.

```
SELECT emp_id, emp_name, salary
FROM employees
WHERE ABS(
    salary - (SELECT AVG(salary) FROM employees)
) = (
    SELECT MIN(
        ABS(salary - (SELECT AVG(salary) FROM employees))
    )
    FROM employees
);
```

15.

```
SELECT dept_id
FROM employees e
GROUP BY dept_id
HAVING MAX(salary) > ALL (
    SELECT salary
```

```
FROM employees m
WHERE m.dept_id = e.dept_id
AND EXISTS (
    SELECT 1
    FROM employees r
    WHERE r.manager_id = m.emp_id
)
);
```